

Assignment (1)

1.	a)	List four access technologies. Classify each one as home access, enterprise access, or wide-area wireless access.	0
	b)	What access network technologies would be most suitable for providing internet access in rural areas?	3
	c)	HFC, DSL, and FTTH are all used for residential access. For each of these access technologies, provide a range of transmission rates and comment on whether the transmission rate is shared or dedicated.	0 3
2.		Suppose users share a 2 Mbps link. Also suppose each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20 percent of the time. (See the discussion of statistical multiplexing in Section 1.3.)	1 0
	a)	When circuit switching is used, how many users can be supported?	
	b)	For the remainder of this problem, suppose packet switching is used. Why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time? Why will there be a queuing delay if three users transmit at the same time?	
	c)	Find the probability that a given user is transmitting.	
	d)	Suppose now there are three users. Find the probability that at any given time, all three users are transmitting simultaneously. Find the fraction of time during which the queue grows.	
3.		Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps.	1 0
	a)	Assuming no other traffic in the network, what is the throughput for the file transfer?	
	b)	Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?	
	c)	Repeat (a) and (b), but now with R_2 reduced to 100 kbps.	
4.	a)	What is self-replicating malware?	0 2
	b)	Describe how a botnet can be created and how it can be used for a DDoS attack.	0 3
	c)	Suppose Alice and Bob are sending packets to each other over a computer network. Suppose Trudy positions herself in the network so that she can capture all the packets sent by Alice and send whatever she wants to Bob; she can also capture all the packets sent by Bob and send whatever she wants to Alice. List some of the malicious things Trudy can do from this position.	0 5

5.		Consider an application that transmits data at a steady rate (for example, the sender generates an N-bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:	1 0
	a)	Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?	
	b)	Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why?	
6.		Review the car-caravan analogy in Section 1.4. Assume a propagation speed of 100 km/hour.	1 0
	a)	Suppose the caravan travels 175 km, beginning in front of one tollbooth, passing through a second tollbooth, and finishing just after a third tollbooth. What is the end-to-end delay?	
	b)	Repeat (a), now assuming that there are eight cars in the caravan instead of ten.	
7.		This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.	1 0
	a)	Express the propagation delay, d_{prop} , in terms of m and s.	
	b)	Determine the transmission time of the packet, d_{trans} , in terms of L and R.	
	c)	Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.	
	d)	Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{\text{trans}}$, where is the last bit of the packet?	
	e)	Suppose d_{prop} is greater than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?	
	f)	Suppose d_{prop} is less than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?	
	g)	Suppose $s = 2.5 \times 10^8$, $L = 1500$ bytes, and $R = 10$ Mbps. Find the distance m so that d_{prop} equals d_{trans} .	
8.		Suppose two hosts, A and B, are separated by 20,000 kilometers and are connected by a direct link of $R = 5$ Mbps. Suppose the propagation speed over the link is 2.5×10^8 meters/sec.	1 0

	a)	Calculate the bandwidth-delay product, $R \cdot d_{\text{prop}}$.	
	b)	Consider sending a file of 800,000 bits from Host A to Host B. Suppose the file is sent continuously as one large message. What is the maximum number of bits that will be in the link at any given time?	
	c)	Provide an interpretation of the bandwidth-delay product.	
	d)	What is the width (in meters) of a bit in the link? Is it longer than a football field?	
	e)	Derive a general expression for the width of a bit in terms of the propagation speed s , the transmission rate R , and the length of the link m .	
9.	a)	What information is used by a process running on one host to identify a process running on another host?	10
	b)	Recall that TCP can be enhanced with TLS to provide process-to-process security services, including encryption. Does TLS operate at the transport layer or the application layer? If the application developer wants TCP to be enhanced with TLS, what does the developer have to do?	
	c)	What is meant by a handshaking protocol?	
	d)	What does a stateless protocol mean? Is IMAP stateless? What about SMTP?	
	e)	How can websites keep track of users? Do they always need to use cookies?	
10.	a)	Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?	10
	b)	Suppose Alice, with a Web-based e-mail account (such as Hotmail or Gmail), sends a message to Bob, who accesses his mail from his mail server using IMAP. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts.	
	c)	What is the HOL blocking issue in HTTP/1.1? How does HTTP/2 attempt to solve it?	
	d)	What is the difference between recursive and iterative DNS queries?	